

WAYFINDER : A Real-Time GPS Based Bus Tracking and Smart Bus Pass Verification System

Work Allocation Report

Project-Based Placement Readiness Programme
Training and Placement Cell

Team Number: VI | Team Name/Repository: team6_wayfinder

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Submission Phase: Phase 2 – Requirement Analysis & Feasibility Review

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Abstract

As part of the Project-Based Placement Readiness Programme, this document formalizes the work allocation for the WAYFINDER project. This distribution of responsibilities ensures that each component of Phase 2—specifically the literature survey, requirement analysis, and feasibility study—is assigned to a dedicated owner to facilitate balanced contributions across the team. These defined roles and responsibilities provide a structured framework for project execution and will serve as the primary reference for coordinating task assignments throughout the upcoming development phases.

Keywords—*Work Allocation, Team Roles, Requirement Analysis, Literature Survey, Feasibility Study, Project Management*

1. Introduction

As per the guidelines provided for the submission of Phase 2 by the training and placement cell, every team is supposed to submit a work allocation document for their project that clearly points out the responsibilities of each team member. The main purpose of this work allocation is to ensure that all the major aspects of the project such as requirement analysis, literature survey, feasibility study are properly handled by someone so that it becomes easier to track the development and implementation of the project from planning to development.

The team allocated work for Phase 2 on the basis of the skills and area of interest of team members identified in Phase 1. In Phase 2, two team members were working on research aspects (literature survey and feasibility study), one team member was handling requirement specification, and one team member was working on applying the concept to the real world by initially developing a rough design of the system that the whole team can agree on for developing a common vision of the project before actually building it.

2. Team Composition

Name	Role in Team
Wesly Philip	Requirement Analyst
Arun P. S	System Designer & Coordinator
Isha Shajahan	Research & Feasibility Analyst
Vyshnav M. R	Research & Feasibility Analyst

3. Detailed Work Allocation

3.1 Literature Survey and Feasibility Study – Isha Shajahan, Vyshnav M. R

Isha Shajahan and Vyshnav M. R were jointly responsible for the research-oriented deliverables of Phase 2. This included conducting a structured literature survey of existing GPS-based bus tracking systems and digital bus pass solutions, reviewing relevant research papers and existing commercial/academic solutions, identifying the technologies used in comparable systems, and documenting the limitations of existing systems. Based on this review, they identified the research gap that the proposed WAYFINDER addresses and explained how the proposed project improves upon existing work. This survey forms part of Requirement Analysis and Literature Survey.

The same members also prepared Feasibility_Study.pdf, evaluating the practicality and viability of the proposed project from four perspectives: Technical Feasibility (availability and suitability of the proposed technology stack), Economic Feasibility (cost of development using free-tier and open-source tools), Operational Feasibility (ease of adoption by drivers, students, and staff), and Schedule Feasibility (achievability of the proposed timeline within the programme's phase deadlines).

3.2 Requirement Analysis – Wesly Philip

Wesly Philip was responsible for the requirement specification component of Phase 2. This involved defining the Functional Requirements of the system – the core features the Driver App, Student/Staff App, and Backend Server are expected to perform, such as live GPS capture, ETA computation, push notifications, and digital bus pass validation – as well as the Non-Functional Requirements, covering quality attributes such as performance, security, reliability, scalability, usability, and maintainability. This work forms the Requirement Analysis section of Requirement_Analysis_and_Literature_Survey.

3.3 System Design and Conceptual Prototyping – Arun P. S

Arun P. S was responsible for coordinating the overall work allocation among team members and for translating the Phase 1 proposal into an early visual and functional design of the system. This included creating conceptual UI/UX mockups and screen layouts for both the Driver App and the Student/Staff App – covering the live tracking map, ETA display, notification flow, and the color-coded digital bus pass – so that the team could arrive at a clear, shared, and concrete understanding of what the final application should look and function like before development begins. This design work directly informed the requirement analysis and feasibility discussions and will serve as a visual reference during the development phase.

4. Work Allocation Summary

Deliverable / Task	Assigned To	Description
Literature Survey	Isha Shajahan, Vyshnav M. R	Review of existing research, solutions, technologies, limitation search gap identification.
Feasibility Study (04_Feasibility_Study.pdf)	Isha Shajahan, Vyshnav M. R	Technical, Economic, Operational, and Schedule feasibility analysis.
Requirement Analysis (Functional & Non-Functional)	Wesly Philip	Definition of core system functionalities and quality attributes.
System Design & Conceptual Prototyping	Arun P. S	UI/UX mockups and screen design for Driver and Student/Staff apps to guide understanding of the system.
Work Allocation Coordination & Compilation	Arun P. S	Coordination of task distribution and compilation of Phase 2 documents.

5. Coordination and Review Process

All team members reviewed each other's sections before compilation to ensure consistency between the requirement analysis, literature survey, feasibility study, and the conceptual system design. Regular updates were committed to the team's GitHub repository (Team_College Bus Tracker) under each member's individual account, in line with the programme's submission process for Phase 2.

6. Schedule

All Phase 2 tasks were completed within the submission window of 11/07/2026 to 20/07/2026, with the literature survey and feasibility study, requirement analysis, and conceptual design carried out in parallel by the respective owners to keep the overall schedule on track.

7. Conclusion

This work allocation reflects a balanced division of Phase 2 responsibilities across the team: research and feasibility analysis by Isha Shajahan and Vyshnav M. R, requirement specification by Wesly Philip, and system design, conceptual prototyping, and coordination by Arun P. S. This division of labour is expected to continue, with role adjustments as needed, into the design and development phases of the College Bus Tracker project.